Roth IRA Model (Pre-Taxed)

Definitions

- Δt: compound time increment (in years)
- $n = 1/\Delta t$: compound frequency (per year)
- r: nominal interest rate (in % per year)
- i: effective inflation rate (in % per year)
- R: real interest rate (in % per year)
- S: total real investment/savings (in \$)
- S_N: total nominal investment/savings (in \$)
- P: principal investment (in \$)
- C: real contribution rate (in \$ per year)
- C_N: nominal contribution rate (in \$ per year)

The Fisher Equation

The real interest rate R is related to r and i as such:

$$1 + R\Delta t = \frac{1 + r\Delta t}{1 + i\Delta t} \implies R = \frac{r - i}{1 + i/n} \tag{1}$$

Nominal Evolution

Over each time increment Δt, the nominal investment changes according to:

$$n\Delta S_N = rS_N + C_N \implies S_N(t + \Delta t) = (1 + {}^{r}\!/\!{}_{n})\, S_N(t) + C_N/n \tag{2}$$

where

- $n\Delta S_N$ is the rate at which S_N changes over the period Δt
- rS_N is the nominal interest gain per Δt

Real Evolution

Over each time increment Δt, the real investment changes according to:

$$n\Delta S = RS + C \implies S(t + \Delta t) = (1 + {}^{R}\!/\!{}_{n})\, S(t) + C/n \tag{3}$$

where

- $n\Delta S$ is the rate at which S changes over the period Δt
- RS is the real interest gain per Δt

The Real Investment

The total real investment is the related to the total nominal investment by the inflationary factor as such:

$$S(t) = \frac{S_N(t)}{\prod_{k=0}^{\lfloor nt \rfloor - 1} (1 + i({}^{k}\!/\!{}_{n})/n)} \tag{4}$$

The Real Contribution

By combining equations (1), (2), (3), and (4), it may be deduced that the real contribution rate is related to the nominal contribution rate by the inflationary factor as such:

$$C(t) = \frac{C_N(t)}{\prod_{k=0}^{\lfloor nt \rfloor} (1 + i({}^{k}\!/\!{}_{n})/n)} \tag{5}$$

The Real Return on Investment

The real return on investment RROI will quantify the real investment growth as such:

$$\texttt{RROI}(t) = \frac{S(t)}{P + \sum_{k=0}^{\lfloor nt \rfloor - 1} C(k/n)/n} - 1 \tag{6}$$

The Nominal Return on Investment

The nominal real return on investment NROI will quantify the nominal investment growth as such:

$$\texttt{NROI}(t) = \frac{S_N(t)}{P + \sum_{k=0}^{\lfloor nt \rfloor - 1} C_N(k/n)/n} - 1 \tag{7}$$

Idealized Inflation

By modeling the inflation rate as constant the real investment and contribution rate simplify as follows:

$$S(t) = \frac{S_N(t)}{(1 + i/n)^{nt}} \qquad \text{and} \qquad C(t) = \frac{C_N(t)}{(1 + i/n)^{nt+1}} \tag{8}$$

Idealized Fixed Nominal Contribution Rate Solution

By solving equations (2) and (3), taking the nominal interest, inflation, and nominal contribution rates to be fixed, given the principal P, it follows that:

$$S(t) = \left(P + \frac{C_N}{r}\right)\left(1 + \frac{R}{n}\right)^{nt} - \frac{C_N}{r}\left(1 + \frac{i}{n}\right)^{-nt} \tag{9a}$$

$$S_N(t) = \left(P + \frac{C_N}{r}\right)\left(1 + \frac{r}{n}\right)^{nt} - \frac{C_N}{r} \tag{9b}$$

Idealized Fixed Nominal RROI & NROI

Applying the idealized fixed nominal contribution rate solution to the RROI and NROI formulas yields:

$$\texttt{RROI}(t) = \frac{\left(P + \frac{C_N}{r}\right)\left(1 + \frac{r}{n}\right)^{nt} - \frac{C_N}{r}}{\left(P + \frac{C_N}{i}\right)\left(1 + \frac{i}{n}\right)^{nt} - \frac{C_N}{i}} - 1 \tag{10a}$$

$$\texttt{NROI}(t) = \frac{\left(P + \frac{C_N}{r}\right)\left(1 + \frac{r}{n}\right)^{nt} - \frac{C_N}{r}}{P + C_N t} - 1 \tag{10b}$$

Generalized Solution

Without fixing any of the parameters, the solution may be generalized as follows:

Take note of the correspondence $(S, C, i) \leftrightarrow (S_N, C_N, 0)$

$$S(t) = P \prod_{k=0}^{\lfloor nt \rfloor - 1} \frac{n + r(k/n)}{n + i(k/n)} + \frac{1}{n} \sum_{k=0}^{\lfloor nt \rfloor - 1} C(k/n) \prod_{m=k+1}^{\lfloor nt \rfloor - 1} \frac{n + r(m/n)}{n + i(m/n)} \tag{11}$$